

# Journal Supplementary I — Relating topics to labels and covariates: a visual annex

Companion to: *Beyond Accuracy* (cross-section relational visualisations)

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2026

## A.1 Scope and motivation

The journal main paper and its earlier supplements (A–H) defend the twelve-axis evaluation framework F-1..F-12 with figures that are either *per-axis scalars* (paired ARI, coherence, seed stability, etc.) or *per-topic intrinsic* (basis spectra, pairwise distances). What is left under-visualised is the **relational** question that motivates topic modelling in the first place: *do the topics actually relate to the labels and covariates we already have?* This supplement fills that gap with six figures, each chosen for a distinct cognitive question, with references to the underlying visualisation tradition.

## B.2 F-7 viewed relationally: topic $\times$ class soft heatmap

This is the probabilistic generalisation of the confusion matrix. Each cell exposes *conditional label support*: a class-pure topic has one tall cell, a mixed topic spreads across columns. The top strip lets the reader gauge whether topic-weighted marginals match the unconditional class prior. Following the mosaic-display tradition of Friendly [1] and the single-cell-LDA precedent of TITAN’s HeatmapTopic.

## C.3 F-7 again: per-topic class-distribution bars

The bars and the heatmap are duals; the heatmap optimises for finding outlier cells, the bars optimise for reading the within-topic distribution shape. Both ship.

## D.4 Topic $\rightarrow$ class Sankey / alluvial flow

References: Sankey diagram tradition; Data-to-Viz Sankey-vs-alluvial distinction [4]; modern `d3-sankey` and `ggsankeyfier` implementations.

## E.5 $\theta$ -PCA scatter coloured by topic / by label

Two-row design follows the `corner.py` scatterplot-matrix convention of Foreman-Mackey [5];  $2\sigma$  ellipses computed via the standard matplotlib gallery recipe [6].

## F.6 Per-topic confidence ridge

The ridge plot is the appropriate idiom for “one continuous distribution per category”. We follow Wilke’s `ggridges` naming [2]; the precursor violin-plot family is Hintze and Nelson [3].

## G.7 HIDSAG: topic $\times$ covariate tag

The continuous mineralogical measurements (Cu rec, Lime cons, mineral abundances per subset) live in the raw HDF5 source files of the figshare collection [8] rather than in the derived JSON exposed by our companion code repository. A future extension could plug those into the same topic-covariate frame using ridge plots per topic on the continuous variable, following the same idiom as Figure 5.

## H.8 HIDSAG continuous-measurement ridges, corner plot, mosaic

After the initial release of this supplement, the HIDSAG continuous mineralogical / geometallurgical measurements were extracted from the per-sample `metadata.json` files in the figshare collection [8] (closing issue `CAOS_LDA_HSI_Paper#2`). A new derived layer at `data/derived/hidsag_topic_measurements/` joins the per-document  $\theta$  matrix with the assay values; 427 records across the five subsets with 5–33 continuous variables per subset.

## I.9 Extended HIDSAG corner plots: MINERAL2 + PORPHYRY

## J.10 PORPHYRY hammock plot — topic Bt% Mb% Py%

## K.11 Cross-method top-word $\lambda$ comparison: LDA / ProdLDA / ETM

## L.12 Why these thirteen figures

The thirteen figures answer thirteen distinct cognitive questions about the topic-label and topic-measurement relationships:

1. Heatmap (Fig. 1) — “which topic-class cells carry mass?”
2. Per-topic bars (Fig. 2) — “what is the within-topic class distribution shape?”
3. Sankey (Fig. 3) — “where does mass flow between topic and class?”
4.  $\theta$  scatter (Fig. 4) — “do topics and labels align in the  $\theta$ -PCA manifold?”
5. Confidence ridge (Fig. 5) — “how sharp is each topic’s posterior?”
6. HIDSAG covariate (Fig. 6) — “do topics carry information about the HIDSAG covariate tags?”
7. HIDSAG ridges (Fig. 7) — “does the dominant-topic assignment separate the continuous mineralogical assays?”
8. HIDSAG corner plot (Fig. 8) — “do topics occupy distinct manifolds in the GEOMET assay space?”
9. HIDSAG mosaic (Fig. 9) — “does the topic membership stochastically order each continuous variable?”
10. MINERAL2 corner (Fig. 10) — “does the high-sulfidation epithermal mineralogy separate by topic?”
11. PORPHYRY corner (Fig. 11) — “does the porphyry-Cu signature separate by topic?”
12. PORPHYRY hammock (Fig. 12) — “how does the joint distribution over topic + 3 assays decompose?”

13. Cross-method words (Fig. 13) — “do LDA / ProdLDA / ETM emphasise the same spectral bands?”

Each chosen idiom dominates its question on the cognitive-ergonomic trade-off between density (information per pixel) and decoding effort (steps the reader must take to extract a claim). See the research note in the repository `supplementary/journal/research-notes/topic-relational-viz.md` for the full literature survey that informed the selection.

## References

- [1] M. Friendly, “Mosaic displays for multi-way contingency tables,” *Journal of the American Statistical Association*, vol. 89, no. 425, pp. 190–200, 1994.
- [2] C. O. Wilke, “ggribes: Ridgeline plots in ggplot2,” R package, <https://wilkelab.org/ggribes/>, 2017, companion blog post “Goodbye joyplots” (2017-09-15) explains the rename from joy plot.
- [3] J. L. Hintze and R. D. Nelson, “Violin plots: A box plot–density trace synergism,” *The American Statistician*, vol. 52, no. 2, pp. 181–184, 1998.
- [4] Data-to-Viz, “The sankey diagram,” <https://www.data-to-viz.com/graph/sankey.html>, 2024.
- [5] D. Foreman-Mackey, “corner.py: Scatterplot matrices in Python,” *Journal of Open Source Software*, vol. 1, no. 2, p. 24, 2016.
- [6] Matplotlib Developers, “Plot a confidence ellipse of a two-dimensional dataset,” Matplotlib gallery recipe, [https://matplotlib.org/stable/gallery/statistics/confidence\\_ellipse.html](https://matplotlib.org/stable/gallery/statistics/confidence_ellipse.html), 2024.
- [7] A. Ehrenfeld, Á. F. Egaña, F. Santibañez-Leal, F. Garrido, M. Ojeda, B. Townley, and F. Navarro, “HIDSAG: Hyperspectral image database for supervised analysis in geometallurgy,” *Scientific Data*, vol. 10, no. 1, p. 164, 2023. [Online]. Available: <https://doi.org/10.1038/s41597-023-02061-x>
- [8] A. Ehrenfeld, “HIDSAG: Hyperspectral image database for supervised analysis in geometallurgy,” figshare Collection, <https://doi.org/10.6084/m9.figshare.c.5983921.v1>, 2023, data collection accompanying [7]; five thematic subsets (GEOMET, MINERAL1, MINERAL2, GEOCHEM, PORPHYRY). Reference implementation at <https://github.com/alges/hidsag>.
- [9] H. Hofmann, “Mosaic plots and their variants,” in *Handbook of Data Visualization*. Springer Berlin Heidelberg, 2008, pp. 617–642.
- [10] M. Schonlau, “Hammock plot for mixed numerical and categorical variables,” *arXiv preprint arXiv:2506.13630*, 2025. [Online]. Available: <https://arxiv.org/abs/2506.13630>

Topic  $\times$  class soft heatmap —  $P(\text{label} \mid \text{dominant topic})$  per scene. Top strip = label marginal weighted by topic prevalence.

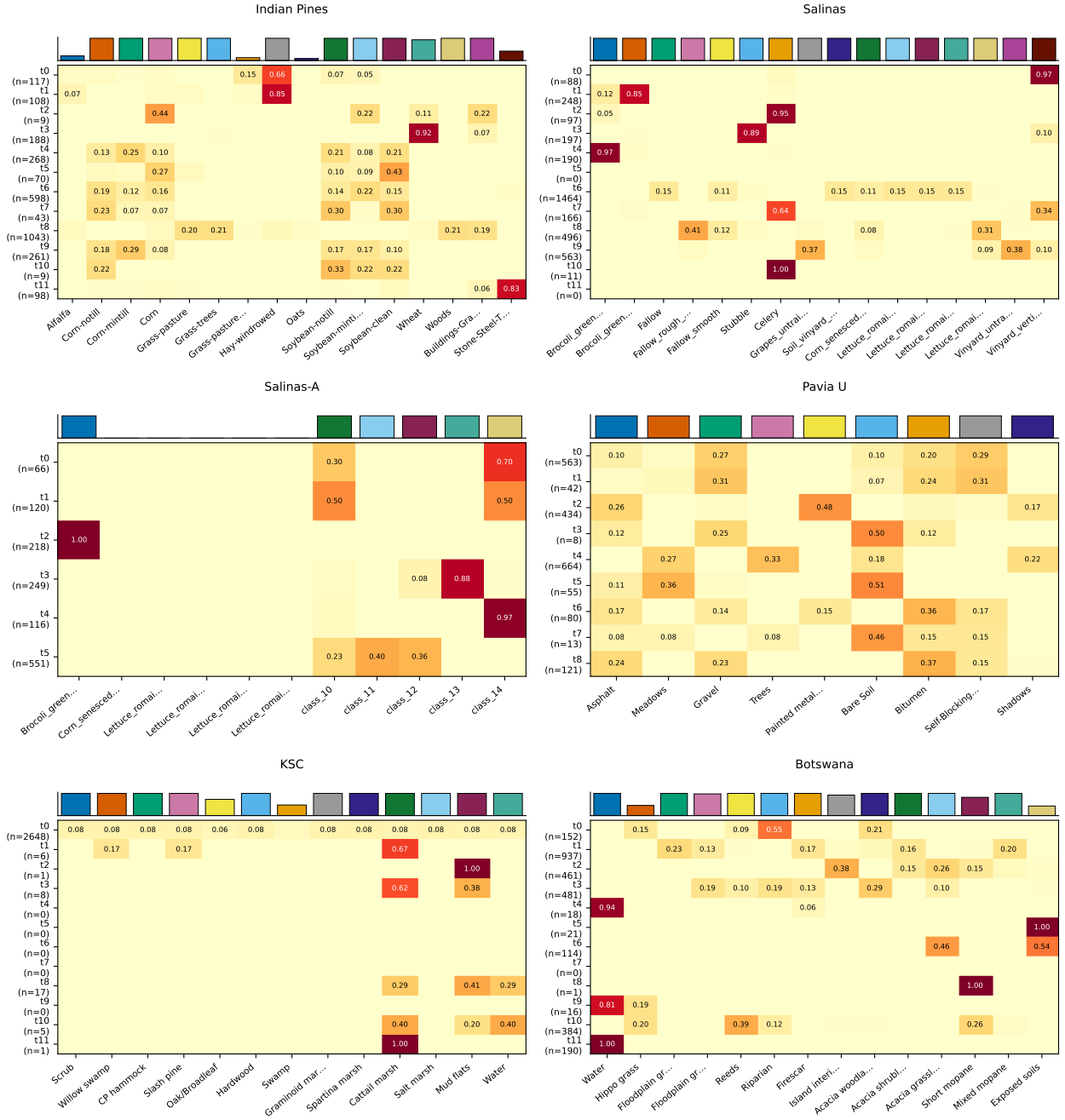


Figure 1: Topic  $\times$  class soft heatmap per labelled scene. Rows = canonical topics (with document count); columns = ground-truth labels; cell intensity =  $P(\ell \mid t)$ , the conditional label distribution given dominant topic. Top strip = label marginal weighted by topic prevalence. Truncated class names appear on the x-axis.

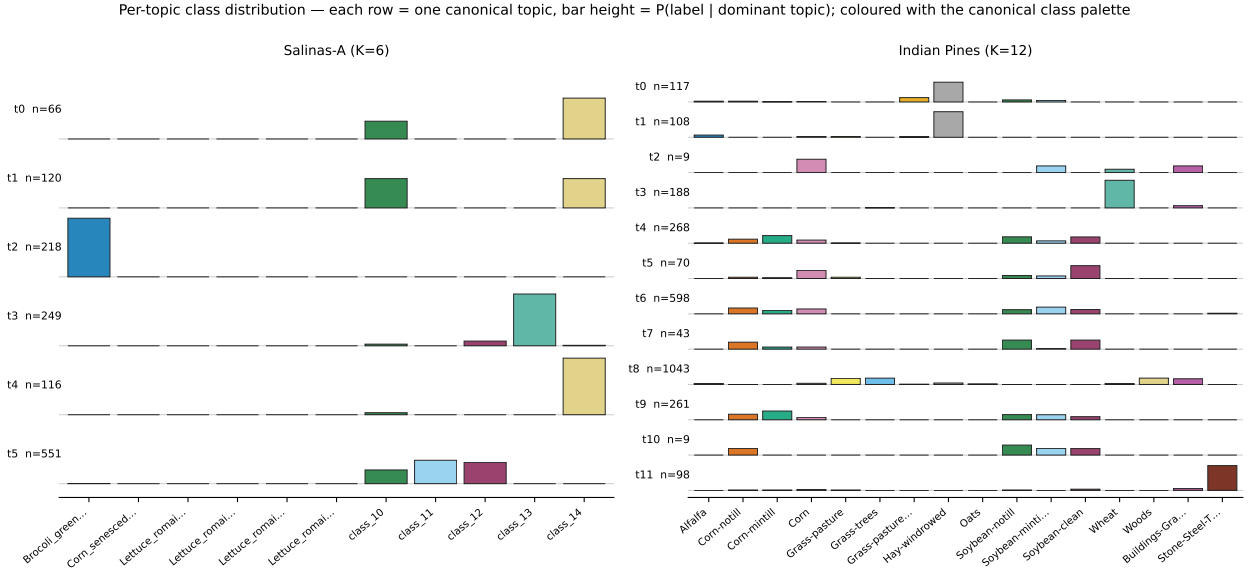


Figure 2: Per-topic class distribution. Each row inside a panel is one canonical topic; bar height =  $P(\ell \mid t)$ , coloured with the canonical class palette so the same class is visually identifiable across topics. A topic that resolves one or two classes shows as a single tall bar; a topic that mixes classes shows as a spread. The categorical analogue of the `ggribes` joy-plot pattern [2,3].

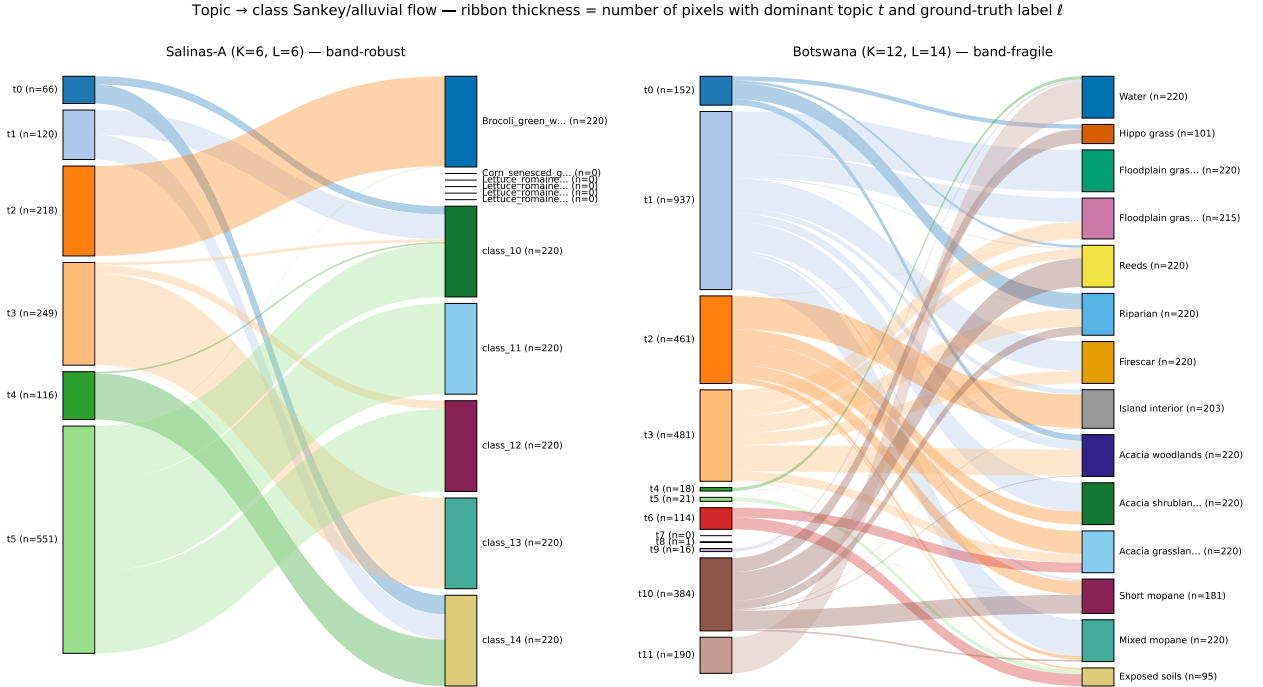


Figure 3: Topic  $\rightarrow$  class Sankey/alluvial flow on Salinas-A (left) and Botswana (right). Left axis nodes = topics; right axis nodes = classes; ribbon width = pixel count. Salinas-A (K=6, L=6, band-robust) shows clean alignment with thin cross-flow. Botswana (K=12, L=14, band-fragile) shows the diffuse coupling that the F-5 paired ARI  $\approx 0.01$  predicts.

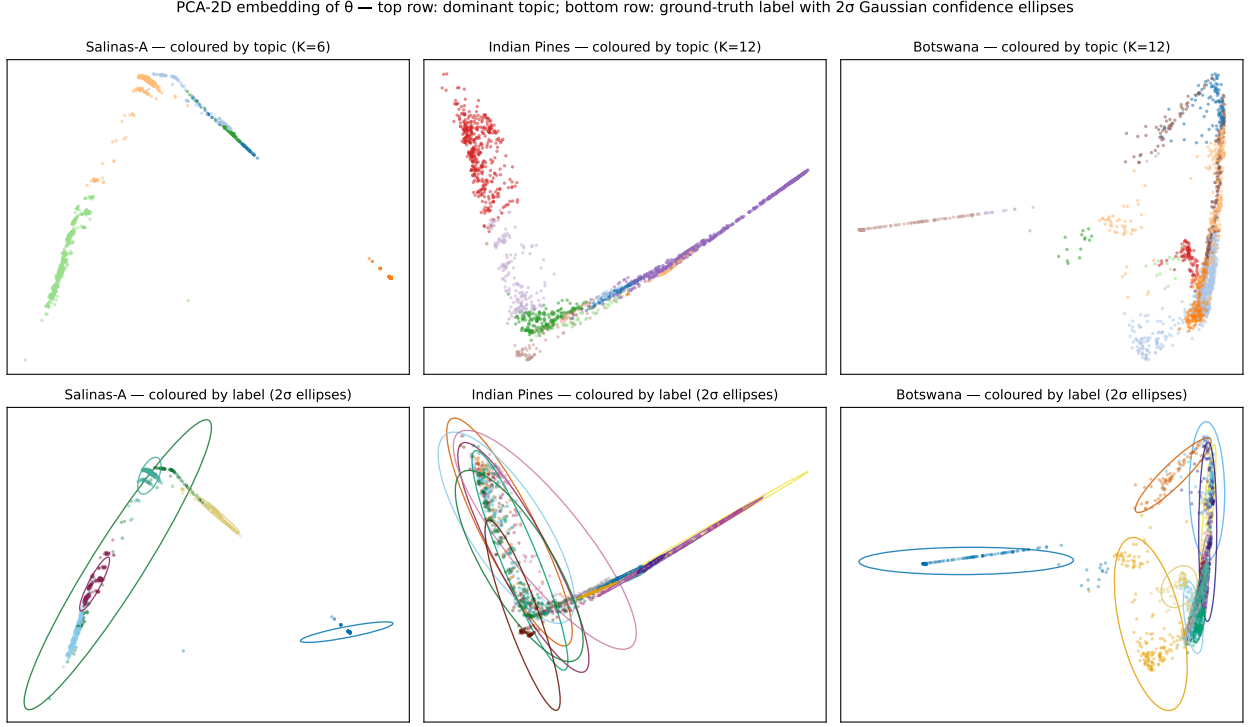


Figure 4: PCA-2D embedding of the per-document topic posterior  $\theta_d$  on three scenes: top row coloured by dominant topic, bottom row coloured by ground-truth label with  $2\text{-}\sigma$  Gaussian confidence ellipses per class. Salinas-A (band-robust): topic and label clusters align tightly. Indian Pines (partial): topic clusters visible but ellipses overlap. Botswana (band-fragile): topic clusters are visible but ground-truth-label ellipses span the whole embedding, exposing the topic–label decoupling.

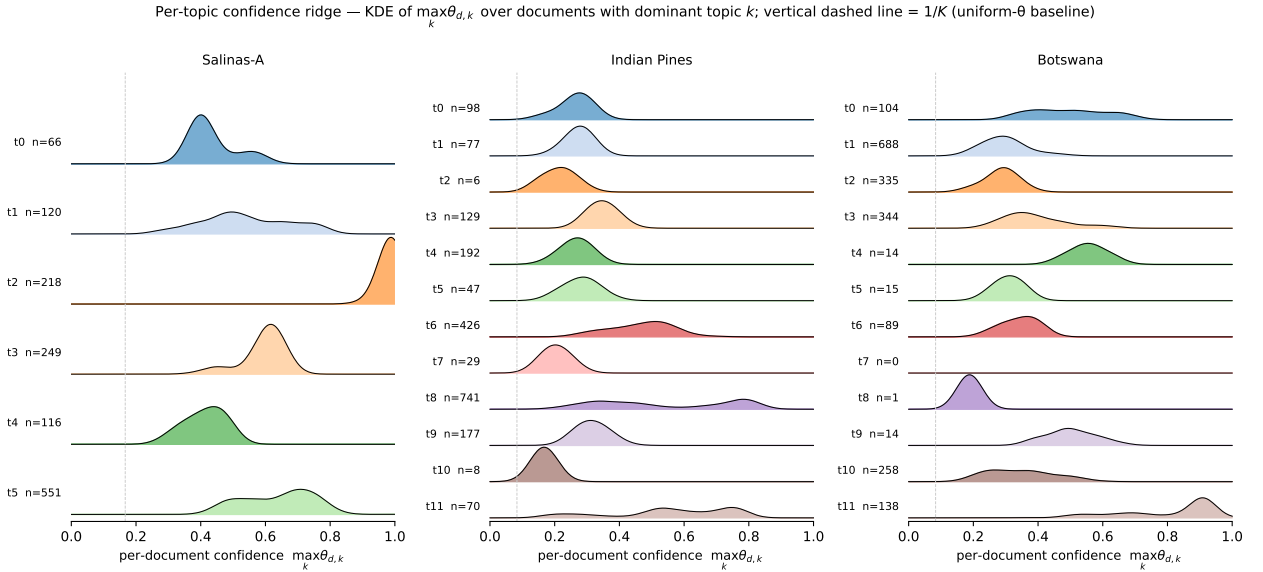


Figure 5: Per-topic KDE of the per-document confidence  $\max_k \theta_{d,k}$  over documents with dominant topic  $k$ . Each topic gets one ridge curve on a shared  $x$ -axis  $[0, 1]$ ; vertical dashed line at  $1/K =$  uniform- $\theta$  baseline. Right-shifted curves = sharp topics; left-shifted = diffuse / collapsed topics.

HIDSAG topic  $\times$  covariate-tag heatmaps — three subsets with non-degenerate tags. GEOMET and MINERAL2 omitted (single 'unknown' covariate, fall-back behaviour from the band-mask builder).

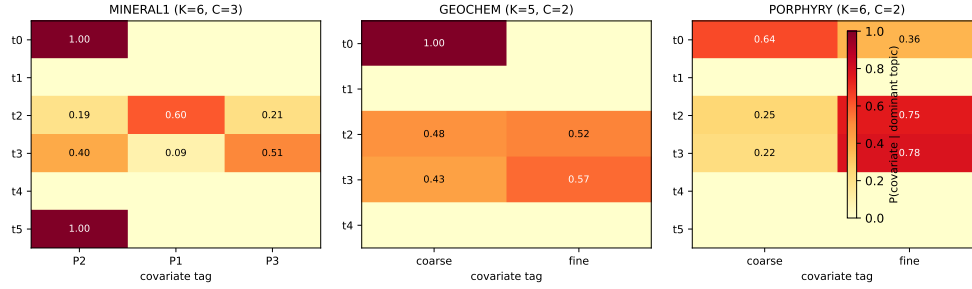


Figure 6: HIDSAG [7,8] topic  $\times$  covariate-tag heatmap on three subsets with non-degenerate covariates (MINERAL1 with P1/P2/P3; GEOCHEM and PORPHYRY with coarse/fine). Cell =  $P(\text{tag} \mid \text{dominant topic})$ . GEOMET and MINERAL2 are omitted because their available covariates collapse to a single **unknown** value at the band-mask builder’s granularity (fall-back behaviour).

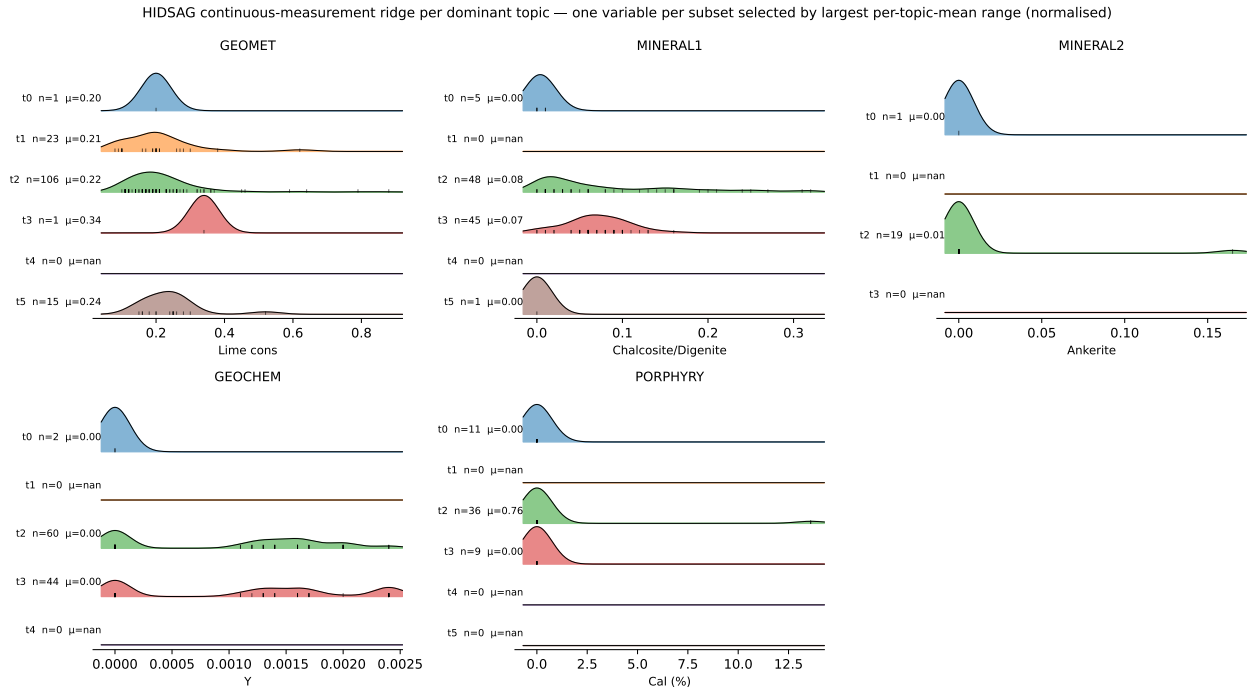


Figure 7: HIDSAG topic-conditional continuous-measurement ridges per subset. For each subset the variable with the largest normalised per-topic-mean range is selected; one KDE ridge per dominant topic on a shared  $x$ -axis (the variable). Per-row  $n$  and mean annotated. Rug marks at the actual observations. The most striking ridge is *PORPHYRY % biotite*: per-topic means range  $1.1 \rightarrow 39.6$ , indicating that the canonical-fit topic basis on PORPHYRY captures the porphyry-alteration mineralogy at a high level of fidelity.

GEOMET corner plot — 5 assays vs dominant topic (K=6; off-diagonal =  $2\sigma$  Gaussian confidence ellipses; diagonal = KDE per topic). Foreman-Mackey 2016 \*corner.py\* idiom.

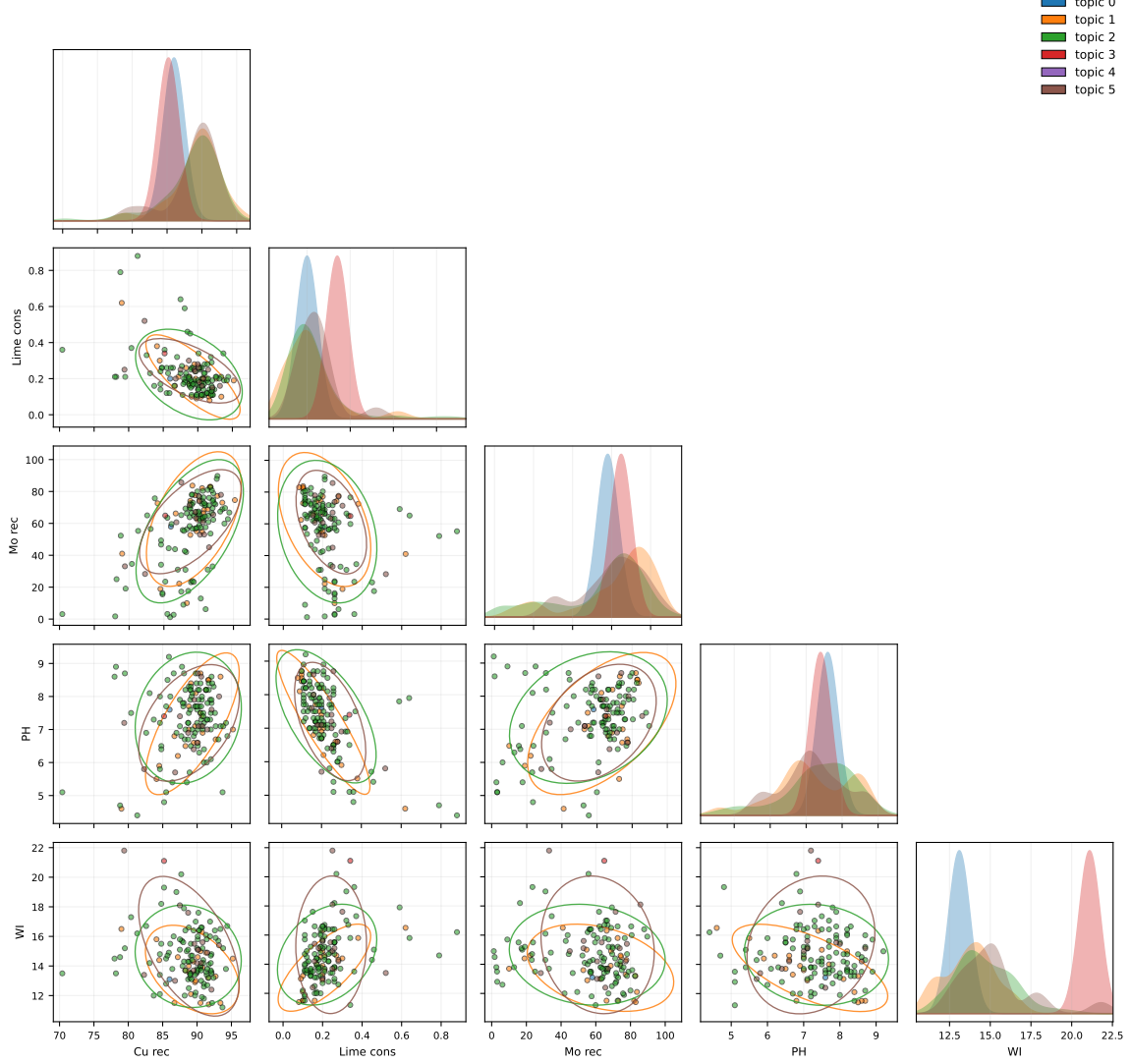


Figure 8: GEOMET corner-plot matrix coloured by dominant topic ( $K = 6$ ). Diagonal: per-variable KDE per topic. Lower triangle: scatter of variable pairs coloured by topic with  $2\sigma$  Gaussian confidence ellipses per topic. Idiom from Foreman-Mackey [5]; ellipses via the canonical matplotlib gallery recipe [6].



HIDSAG mosaic:  $P(\text{quartile-bin} \mid \text{dominant topic})$  on a selected continuous variable per subset. Bins are scene-wide quartiles; rows are normalised so each topic sums to 1.

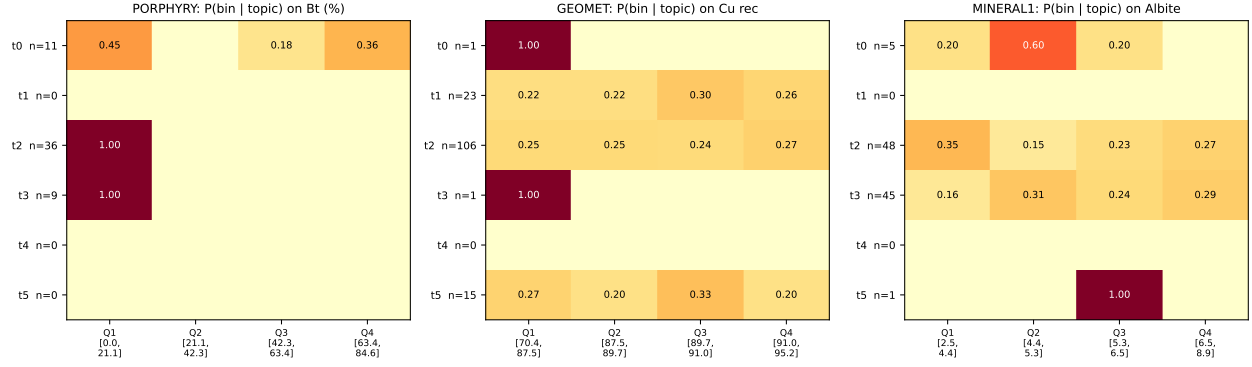
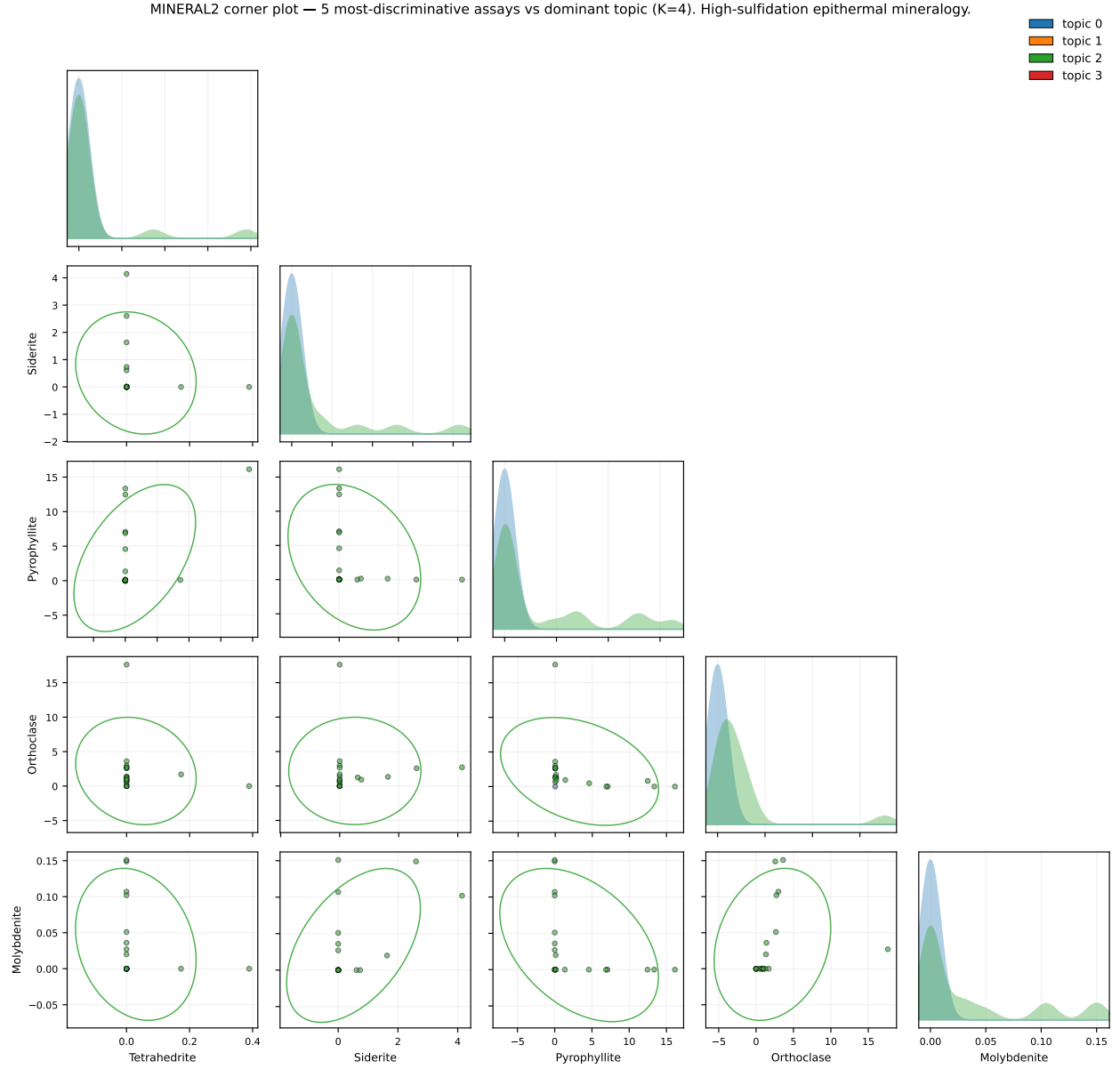


Figure 9: HIDSAG mosaic / fluctuation heatmap of  $P(\text{quartile} \mid \text{dominant topic})$  on three subsets and the variable picked per subset (PORPHYRY: % biotite; GEOMET: Cu recovery; MINERAL1: albite). Bins are scene-wide quartiles; rows normalised. Captures non-monotone topic-vs-quartile patterns that the smooth ridge KDE can hide. Idiom: Hofmann [9]; Friendly [1].



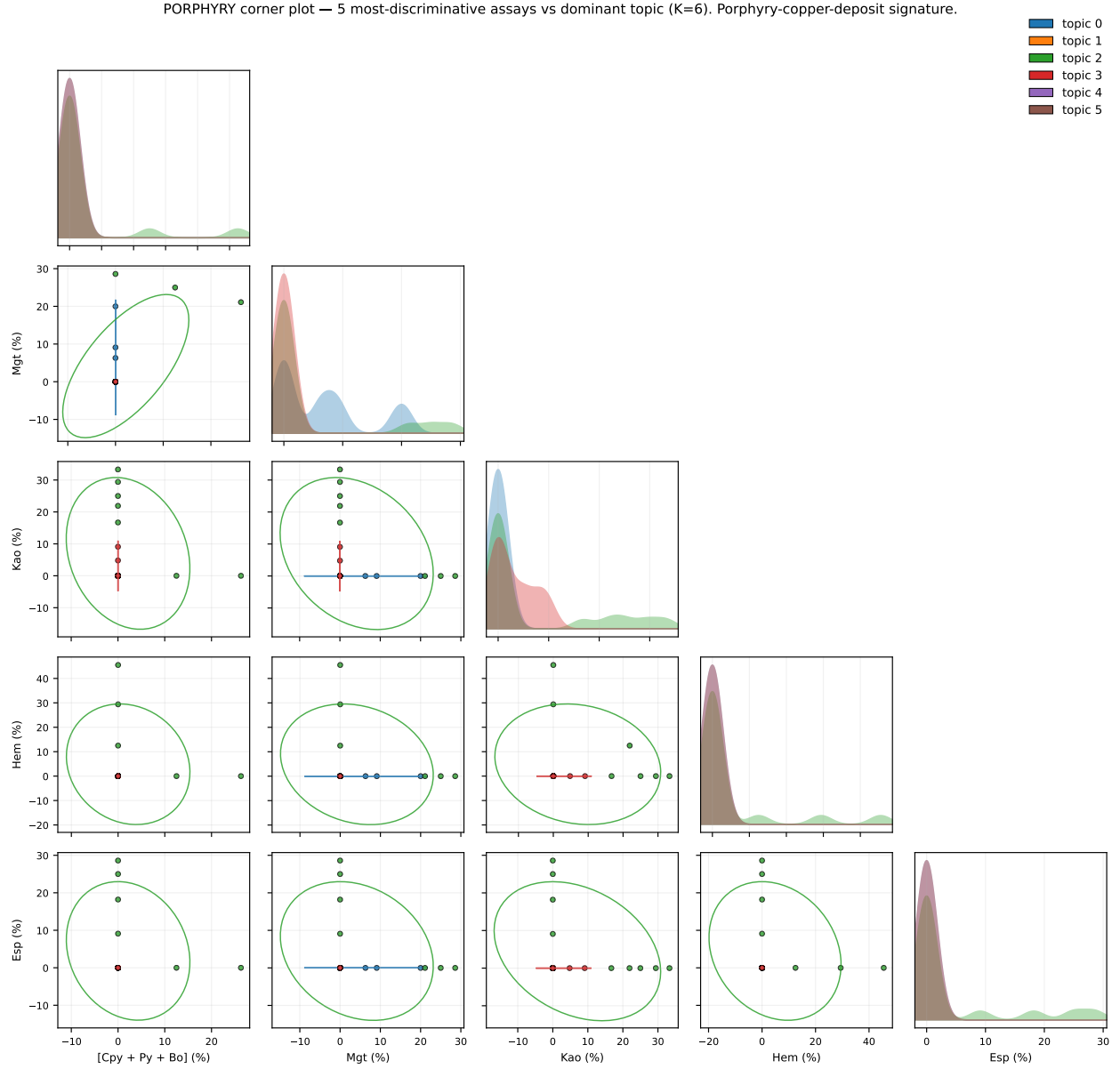


Figure 11: PORPHYRY corner plot ( $K = 6$ ,  $N = 56$ ). Auto-selected top-5 most-discriminative mineral abundances (% biotite, % molybdenite, % quartz, % pyrite, combined sulfide [Cpy+Py+Bo]%). The Bt%-vs-Mb% panel surfaces the biotite-dominant versus molybdenite-dominant topic families directly.

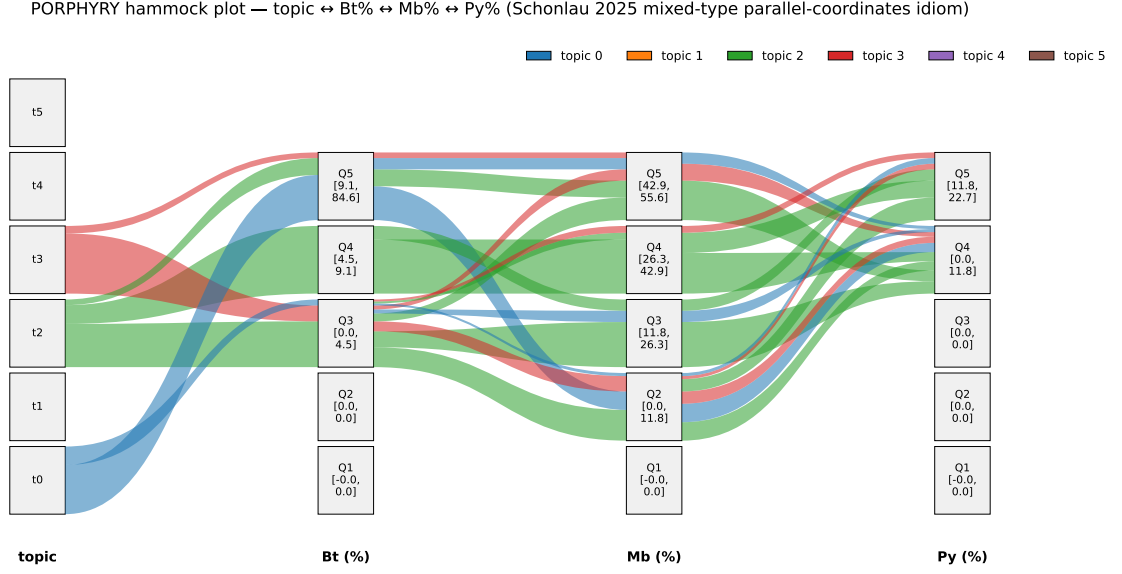


Figure 12: Hammock plot [10] on PORPHYRY combining the categorical *dominant topic* axis ( $K = 6$ ) with three continuous mineralogical assays (% biotite, % molybdenite, % pyrite). Continuous axes are binned into scene-wide quintiles ( $Q_1..Q_5$ ). Ribbon thickness = document count; topic-coloured (tab10). The hammock idiom unifies mosaic plots (categorical-only) and parallel-coordinates (continuous-only) on the same diagram.

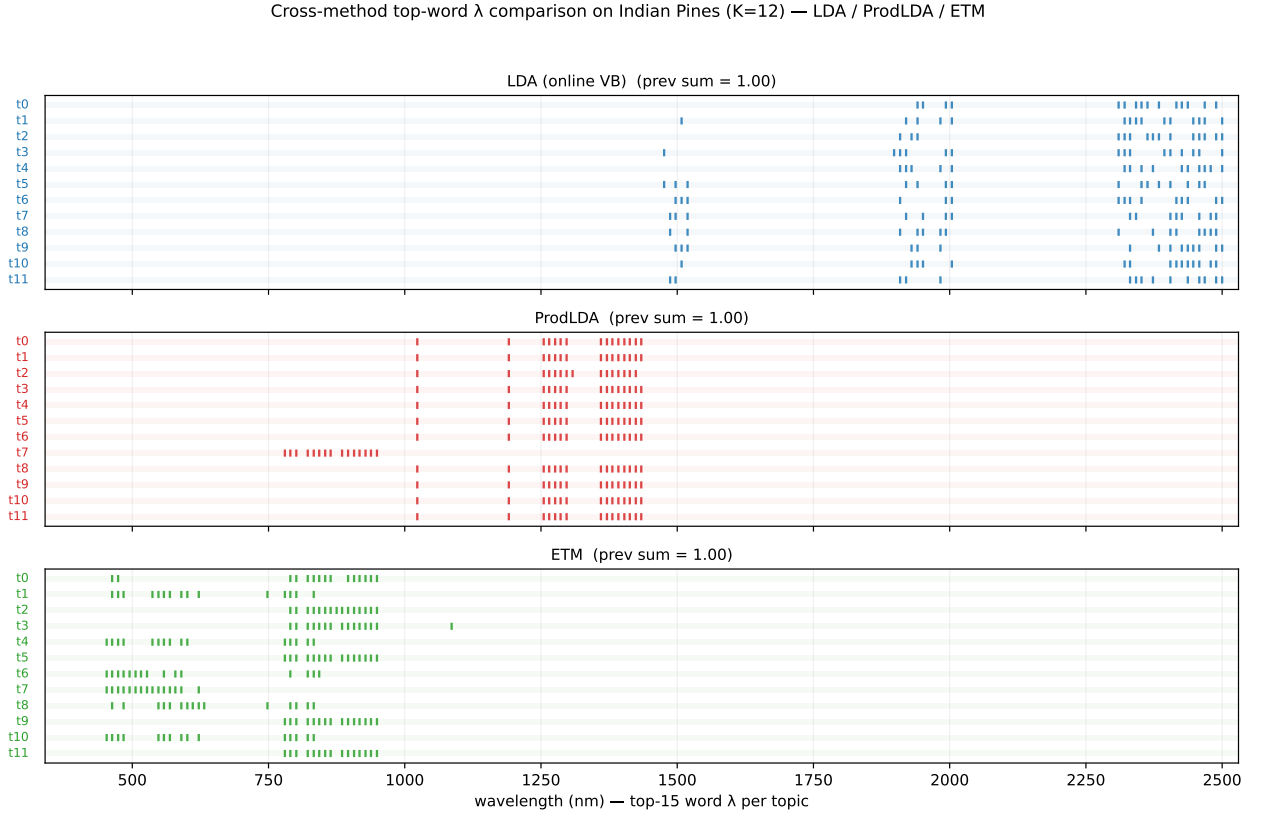


Figure 13: Cross-method top-word  $\lambda$  comparison on Indian Pines ( $K = 12$ ). One panel per method (LDA, ProDLDA, ETM); each panel shows  $K$  rows of top-15-word wavelength tick marks. Complements the coherence-vs-ARI scatter (which reports scalars) with the spectral identity of each method's basis: if LDA and ETM emphasise the same wavelength bands but ProDLDA's emphasis shifts, the shift *is* the interpretability difference.